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**A Power Generation System
Utilizing Space-Time Expansion**

For

Dr. S. Greer

By

[REDACTED]
[REDACTED]

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Technical Proposal

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2.0 Description

The proposed power generation system is based upon the fact that the Lorentz Force in certain conditions can be non-conservative, thus indicating the important relationship to open systems and hence external forces. In this particular application, there is a frame of reference shift between bound ions and conduction electrons in a conducting medium. To create an optimal frame of reference shift, it is important to set up an asymmetry of the electric field and establish a leakage current through a conductor and a dielectric, which also serves to store charge. This will result in a force that can be used in a power generation system.

This can be accomplished with asymmetric capacitors, utilizing corona generated in air as a leaky dielectric and a non-leaky dielectric for storing charge. This combination of dielectrics is straight forward, but ungainly, and results in structural mass that is not part of the resulting force. Utilizing corona as a leaky dielectric is expedient; however, it is very inefficient and consumes a tremendous amount of power. A bifilar pendulum can be used to measure generated force with respect to voltage and current, and a rotor can be used to measure the instantaneous power generated in rotation when balanced against a calibrated drag.

In the proposed effort a pendulum and rotor will be utilized. The performance of the asymmetrical capacitors is directly related to the nature of the materials and geometries that will be employed. For example, purity of materials, surface area with respect to volume, and cross-sectional areas will be investigated. Analytical models will be

constructed to ascertain trends in performance in order to facilitate design of the first-generation structures using the asymmetric capacitive force.

3.0 Technical Objectives

3.1 Background

There are two force laws of motion in electromagnetism: the Ampere Force Law and the Lorentz Law. This leads to the Lorentz-Maxwell equation of motion

$$\frac{d\vec{P}_i}{dt} = \vec{F}_{Lij}$$

Here $\vec{F}_{Lij}$ is the Lorentz Force applied to particle i and is expressed as

$$\vec{F}_{Lij} = q_i(\vec{E}_j + \vec{U}_i \times \vec{B}_j)$$

$\vec{E}_j$ and $\vec{B}_j$ are external fields produced by charged particles j . The velocity $\vec{U}_i$ of particle q_i is relative to an inertial frame, and as such will have different values in different inertial reference frames.

It appears that the Lorentz Force does not satisfy Newton's Third Law, since

$$\vec{F}_{Lij} \neq -\vec{F}_{Lji}$$

This implies that the external forces can perform work whose energy is provided either by the medium or alternate frames of reference. It is important to note that the magnetic part of the Lorentz Force does not perform work on a charged particle.

The definition of an inertial frame implies that this frame has at its origin a particle with infinite mass. The mass of any real particle is finite, and it has been shown that external forces exist that can perform work whose energy is provided by a medium or alternate frames of reference. To show the existence of these alternate frames, examine the interaction between two moving charges with forces that appear to violate Newton's Third Law. Lorentz Forces exerted by freely moving charges on one another are not equal and opposite in principle, and it follows that a system of a pair of charged particles in relative motion can change the motion of its center of mass without external influence. As an example, two charges q_1 and q_2 are moving with velocities U_1 and U_2 , relative to a reference frame where the background frame is at rest. There is no change of reference frame. Charge q_1 exerts a force on charge q_2 as

$$\vec{F}_{21} = q_2(\vec{E}_1 + \vec{U}_2 \times \vec{B}_1)$$

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where $\vec{E}_1$ and $\vec{B}_2$ are the electric and magnetic fields produced by q_1 at the position of q_2 . Also, charge q_2 exerts a force on charge q_1 , as

$$\vec{F}_{12} = q_1(\vec{E}_2 + \vec{U}_1 \times \vec{B}_2)$$

The two forces can in general have different magnitudes and directions, and are expressed as

$$\frac{d\vec{P}_1}{dt} + \frac{d\vec{P}_2}{dt} = \vec{F}_{12} + \vec{F}_{21} \neq 0$$

This can be shown by expressing the Lorentz Force in terms of scalar and vector potentials.

$$\vec{E} = -\vec{\nabla}\phi - \frac{\partial \vec{A}}{\partial t}$$

and

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

Then

$$\vec{F}_{12} = q_1(\vec{\nabla}\phi_2 - \frac{\partial \vec{A}_2}{\partial t} + \vec{U}_1 \times \vec{\nabla} \times \vec{A}_2)$$

$$\vec{F}_{21} = q_2(-\vec{\nabla}\phi_1 - \frac{\partial \vec{A}_1}{\partial t} + \vec{U}_2 \times \vec{\nabla} \times \vec{A}_1)$$

The momentum of the particles is expressed as

$$\vec{P}_1 = m_1 \gamma_1 \vec{U}_1 \quad \vec{P}_2 = m_2 \gamma_2 \vec{U}_2$$

where

$$\gamma_i = (1 - U_i^2 / c^2)^{-\frac{1}{2}}$$

where γ is a relationship of particle velocity with respect to the speed of light. In relativistic mechanics, the equality of action and reaction is not satisfied and thus is not due to the relativity of simultaneity or the retardation effect. As a result, two charged

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particles do not constitute a closed system, thus giving rise to the appearance of the violation of Newton's Third Law by Lorentz Forces. It also follows that the conservation of mechanical energy of the system cannot be verified. The classical definition of the center of mass is the only definition physically meaningful, and yields the following.

$$\begin{aligned} m \frac{d^2 \vec{r}}{dt^2} &= \vec{F}_{12} + \vec{F}_{21} - m_1 \vec{U}_1 \frac{d\gamma_1}{dt} - m_2 \vec{U}_2 \frac{d\gamma_2}{dt} \\ &= F_G \\ &\neq 0 \end{aligned}$$

where F_G is the resulting force.

In terms of the Coulomb gauge, there are no external source terms, so $\vec{\nabla} \phi = 0$, and $\vec{\nabla} \cdot \vec{A} = 0$. Then the forces become

$$\vec{F}_{12} = -q_1 \frac{\partial \vec{A}_2}{\partial t} \quad \vec{F}_{21} = -q_2 \frac{\partial \vec{A}_1}{\partial t}$$

Substituting for F ,

$$q_1 \frac{\partial \vec{A}_2}{\partial t} + q_2 \frac{\partial \vec{A}_1}{\partial t} + m_1 \vec{U}_1 \frac{d\gamma_1}{dt} + m_2 \vec{U}_2 \frac{d\gamma_2}{dt} \neq 0$$

Rearranging,

$$-\frac{d}{dt}(q_1 \vec{A}_2) - \frac{d}{dt}(q_2 \vec{A}_1) \neq \frac{d}{dt}(m_1 \vec{U}_1 \gamma_1) + \frac{d}{dt}(m_2 \vec{U}_2 \gamma_2)$$

It can be seen that field theory attributes momentum to the electromagnetic field to allow a particle to interact only with fields at the position of the particle. This precludes possible instantaneous particle interaction except as an approximation. The interaction between particles proceeds as a transfer of momentum from one particle to the field. Then the field transports the momentum at the speed of light to the position of the second particle, where it can be transferred from the field to the other particle. This is not symmetric. To take the fields into account does not change the fact that there is a resultant motion of the center of mass.

For small accelerations, it is possible to only take into account the velocity fields at the simultaneous positions r_1 and r_2 of the particles. The expression for the electric fields in terms of velocity is given as

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$$\vec{E}_1 = k \frac{q_1}{R^2} \frac{1 - \beta_1^2}{(1 - \beta_1^2 \sin^2 \theta)^{3/2}} \vec{n}$$

$$E_{21} = -k \frac{q_2}{R^2} \frac{1 - \beta_2^2}{(1 - \beta_2^2 \sin^2 \theta)^{3/2}} \vec{n}$$

where $\beta = U/c$ is the particle velocity with respect to the speed of light, $R = r_2 - r_1$,

$$\vec{n} = \frac{\vec{R}}{R}, \text{ and } k = \frac{1}{4\pi\epsilon_0}.$$

The velocities are small with respect to the speed of light ($U \ll c$), so the expressions for the electric fields can be expanded as

$$\vec{E}_1 \approx k \frac{q_1}{c^2 R^2} (c^2 + \frac{1}{2} \vec{U}_1^2 - \frac{3}{2} (\vec{U}_1 \cdot \vec{n})^2) \vec{n}$$

$$\vec{E}_2 \approx -k \frac{q_2}{c^2 R^2} (c^2 + \frac{1}{2} \vec{U}_2^2 - \frac{3}{2} (\vec{U}_2 \cdot \vec{n})^2) \vec{n}$$

Also, the magnetic fields are expressed as

$$\vec{B}_1 = \vec{U}_1 \times \vec{E}_1 \quad \vec{B}_2 = \vec{U}_2 \times \vec{E}_2$$

The Lorentz Force for the particles is

$$\begin{aligned} \vec{F}_{12} &= q_1 (\vec{E}_2 + \vec{U}_1 \times \vec{B}_2) \\ &= q_1 (\vec{E}_2 + \vec{U}_1 \times (\vec{U}_2 \times \vec{E}_2)) \end{aligned}$$

and

$$\begin{aligned} \vec{F}_{21} &= q_2 (\vec{E}_1 + \vec{U}_2 \times \vec{B}_1) \\ &= q_2 (\vec{E}_1 + \vec{U}_2 \times (\vec{U}_1 \times \vec{E}_1)) \end{aligned}$$

The force $\vec{F}_G$ is

$$\vec{F}_G = \vec{F}_{12} + \vec{F}_{21} \neq 0$$

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$$\vec{F}_G = q_1(\vec{E}_2 + \vec{U}_1 \times (\vec{U}_2 \times \vec{E}_2)) + q_2(\vec{E}_1 + U_2 \times (\vec{U}_1 \times \vec{E}_1))$$

Substitute for $\vec{E}_1$ and $\vec{E}_2$, and let $q = q_1 = -q_2$, $\vec{V} = \vec{U}_2 - \vec{U}_1$, and $\vec{U} = U_2$, where V is the relative velocity and U is the absolute velocity. Here it is assumed that $\vec{V} \ll \vec{U}$. The expression for the force becomes

$$F_G = k \frac{2q^2}{c^2 R^2} UV$$

These expressions of the force depend upon an absolute velocity U that is defined with respect to a preferred reference frame. This can be generalized by considering a cluster of N particles that are closed to one another and use the Jacobi coordinates in order to calculate the force for the cluster. It is important to obtain an expression that will depend on the absolute motion $\vec{U}$ of the center of mass of the cluster with respect to a preferred reference frame.

The fabric of spacetime is expanding at a velocity corresponding to reshifted blackbody radiation. For $\beta = U/c < 1$, the temperature measured by a moving observer is

$$T_0 \approx T_s(1 - \beta \cos \theta)$$

where θ is the angle between the direction of motion and the direction of measurement, T_0 is the reference temperature, and T_s is the temperature shift. This corresponds to a velocity of around 400 km/sec.

For an example of the application of the force, consider the following. The expression contains charge q^2 , which can be readily obtained by capacitance, voltage, and participating conduction electrons in the capacitor,

$$q^2 = (cV + q_e)^2$$

A basic capacitor consists of two conducting metal surfaces with a dielectric sandwiched between. The metal surfaces contain conduction electrons, and as the capacitor moves through a background reference frame, these electrons will be accelerated differently, lagging behind with a relative drift velocity $\vec{V}$.

$$\vec{V} = \vec{U}_i - \vec{U}_e$$

where $\vec{U}_i$ is the ion velocity with respect to the background reference frame, and $\vec{U}_e$ is the electron velocity with respect to the background reference frame. Therefore the

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stimulated force will exist, if there is a drift velocity between electrons and the ions. Also, a leakage current (leaky dielectric) will greatly enhance the force. This is due to the fact that the electron drift velocity $\vec{V}$ is related to the current density $\vec{j}$ by $\vec{j} = \rho\vec{V}$, where ρ is the charge density of the material. The geometry of the capacitors can be long and planar, cylindrical, and combinations of planes, cylinders, and spheres. The metal surfaces can be constructed of most readily available metallic materials, such as copper and aluminum. The dielectrics can be made of most plastic based materials and polyamide films. The capacitances will vary from 500 picofarads to a few picofarads. The distances between the metallic surfaces can vary from a few tenths of an inch to almost 12 inches and the corresponding potentials can range from 10,000 volts DC to 100,000 volts DC. The electron drift velocity of most metals is on the order of 0.02 meters/sec and the ions have an absolute velocity of 400,000 m/sec (Earth frame). The key to a large force is based on three requirements. The first is the interaction of all of the conduction electrons in the metallic surfaces. For conductors the number of available electrons is expressed as

$$n = \frac{N_o \rho}{M} \xi V_{vol}$$

where N_o is avagodros number
M is the molecular weight
 ρ is the density
 ξ is the number of free electrons / atoms
 V_{vol} is the volume of the material

For copper $n/V_{vol} \sim 8 \times 10^{22}$ electrons/cm³. Here it can be seen that with available materials, the force F_G can approach 100,000 newtons.

The second requirement is the purity of the materials. As the purity is increased, the drift velocity of the electrons is increased. It is possible to achieve thin film purities that would allow the electrons to become almost ballistic in the conductors, thus increasing the drift velocity $\vec{V}_e$ by almost a factor of 1000.

The third requirement is a dielectric material that has a good dielectric constant and electron mobility. This allows for charge storage and leakage at the same time. Combinations of semiconductors and alloys would satisfy this requirement.

With these requirements working together, it should be possible to achieve a force F_G on the order of 10×10^6 newtons (2×10^6 lbf) in a volume of 1 meter cubed with a voltage of 50,000 DC at a current of 0.001 ampere. It is important , however, to realize material constraints and engineering practicalities in actual designs.

3.2 Technical Objectives

There are four main technical objectives discussed within this proposal. They are:

- Investigate the efficiency of generating the force with respect to conducting material purities and geometries.
- Investigate the efficiency of generating the force with respect to leaky dielectrics and geometries.
- Investigate the ability to scale the force.
- Utilize this technology for energy/power generation applications.

4.0 Work Plan

In order to achieve the objectives of this proposal the project is divided into five major tasks in accordance with a Work Breakdown Structure (WBS).

4.1 Investigation of conducting materials (WBS 1.0)

The conduction electrons in a conductor can be treated as a “gas,” using Fermi-Dirac statistics. The electrons can have a Maxwellian velocity distribution and as such randomly distribute energy throughout a conductor. When an electrical field is placed across a conductor, the conduction electrons drift towards the positive pole and will do so with a spectrum of energies due to thermalization. In principle, the purer the material, the smaller the effect will be. Therefore the electrons can become ballistic in nature, and the resulting drift velocity increases many orders of magnitude. Parametric analysis and tests will be conducted to ascertain how purity can impact this phenomenon. As an example, for a given voltage and geometry, the drift velocity of the conduction electrons will be measured for a variety of conductor purities. Similarly, for a given cross section of conductor, the current will be varied in order to evaluate change in drift velocity. These types of tests will be conducted on a bifilar pendulum and a rotor in order to evaluate the magnitude of the force. Resulting test data will be put into a performance matrix. The resulting analysis will develop a performance model that will include variables such as voltage, current, purity, drift velocity, and geometry.

4.1.1 (WB 1.1) A parametric analysis will be conducted to investigate the interdependency of voltage, geometry, drift velocity, and purity with respect to the force. This will be accomplished by the development of performance models in order to evaluate trends for design and test.

4.1.2 (WBS 1.2) Capacitors will be designed and constructed based on performance models. Test requirements will be generated.

4.1.3 (WBS 1.3) Tests will be conducted utilizing a bifilar pendulum and a rotor in order to observe performance of the capacitors. The results will be compared to the performance models. A performance matrix will be generated.

4.2 Investigation of leaky dielectrics (WBS 2.0)

Generation of the force is related to stored charge and a leakage current. It is ideal to have one material that has a high dielectric constant as well as high electron mobility. Currently the asymmetric capacitors utilize corona to provide a leaky dielectric and basic non-conducting materials for a non-leaky dielectric in order to store charge. This adds mass that does not participate in the generation of the force. A single material that can both store charge and be leaky would be optimal. Parametric analysis and tests will be conducted to investigate material compositions and the resulting impact on dielectric constant, electron mobility, and mass. These types of tests will be conducted on a bifilar pendulum and rotor in order to evaluate the magnitude of the force. Resulting test data will be put into a performance matrix. The resulting analysis will develop an analytical performance model to include variables such as voltage, current, composition, dielectric constant, electron mobility, and geometry.

4.2.1 (WBS 2.1) A parametric analysis will be conducted to investigate the interdependency of composition, dielectric constant, electron mobility, and mass with respect to the force. This will be accomplished by the development of performance models in order to evaluate trends for design and test.

4.2.2 (WBS 2.2) Capacitors will be designed and constructed based on performance models. Test requirements will be generated.

4.2.3 (WBS 2.3) Tests will be conducted utilizing a bifilar pendulum and a rotor in order to observe performance of the capacitors. The results will be compared to the performance models. A performance matrix will be generated.

4.3 Investigation of scalability (WBS 3.0)

The magnitude of the force is related to current density and hence cross sectional area. Ultrapure materials and leaky dielectrics allow structures to be thin, thus minimizing nonessential mass. Parametric analysis and tests will be conducted to investigate the relationship between force, current density, and cross section. These types of tests will be conducted on a bifilar pendulum and rotor in order to evaluate the magnitude of the force. Resulting test data will be put into a performance matrix. The resulting analysis will develop a model to include variables such as cross section, length, voltage, and current.

4.3.1 (WBS 3.1) A parametric analysis will be conducted to investigate the interdependency of current density and cross sectional area with respect to the force. This will be accomplished by the development of performance models in order to evaluate trends for design and test.

4.3.2 (WBS 3.2) Capacitors will be designed and constructed based on performance models. Test requirements will be generated.

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4.3.3 (WBS 3.3) Tests will be conducted utilizing a bifilar pendulum and a rotor in order to observe performance of the capacitors. The results will be compared to the performance models. A performance matrix will be generated.

4.4 Power Generation (WBS 4.0)

In order to produce useful work, the generated force must be applied to technologies that are readily available. This would allow the current infrastructure to be interfaced for power distribution. To accomplish this, the generated force will have to be cyclic in nature either by internal generation or by continually driving a rotating system (flywheel). Parametric analysis and baseline tests will be conducted to investigate the relationships between the generated force and power generation techniques.

4.4.1 (WBS 4.1) A parametric analysis will be conducted to investigate the input requirements for the generated force with respect to the available techniques for power generation, their efficiencies, and the ability to put power back into the input supply. This must be accomplished with a net power gain. Performance models will be developed in order to evaluate trends for design and test.

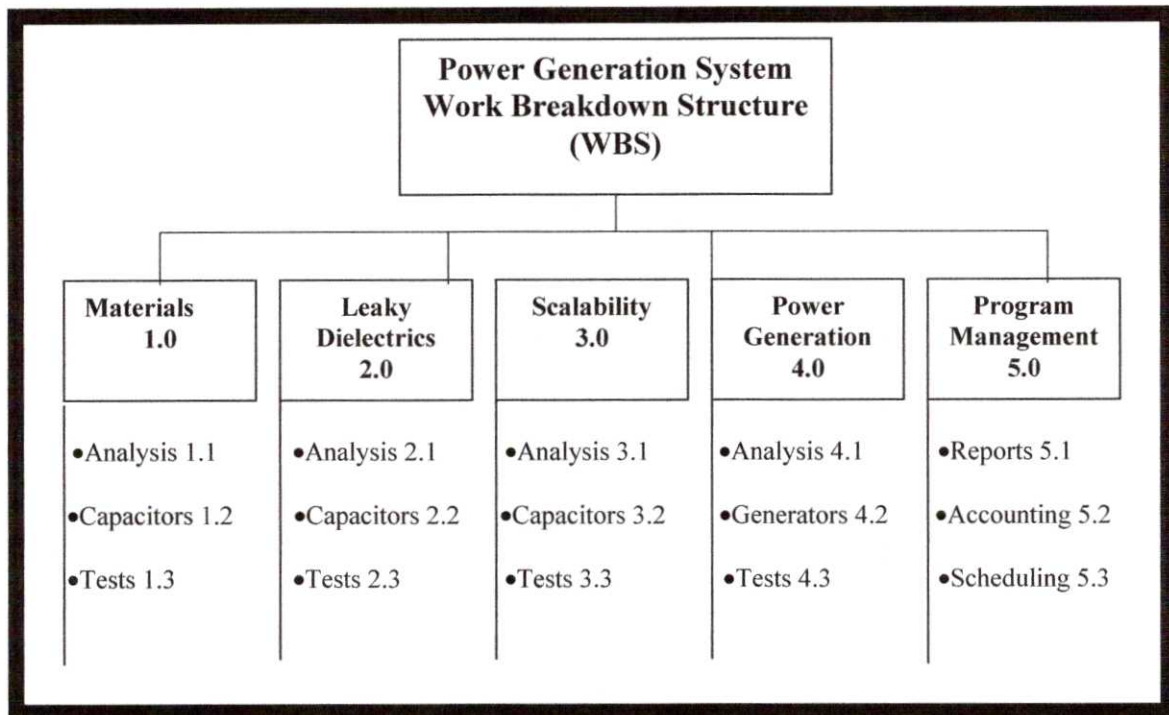
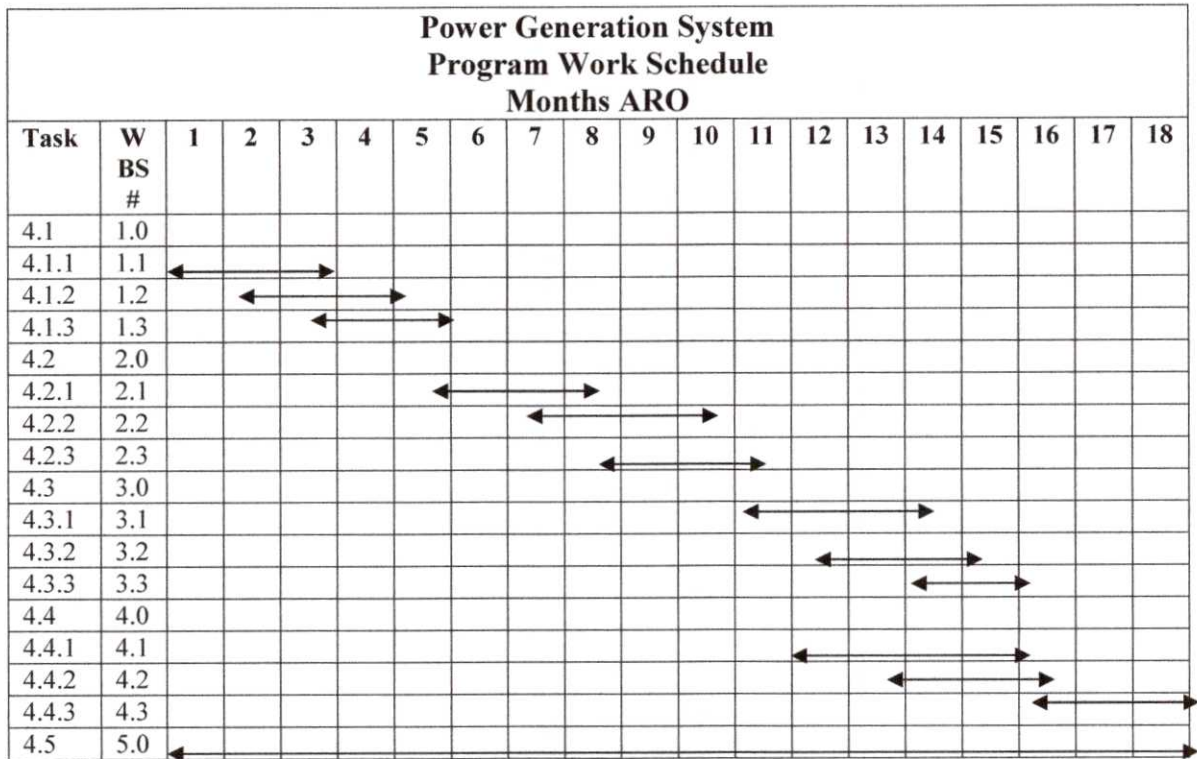
4.4.2 (WBS 4.2) A power generation system consisting of driving supply, power takeoff, and feedback will be designed and constructed, based on performance models. Test requirements will be generated.

4.4.3 (WBS 4.3) Tests will be conducted with existing capacitors coupled with a driving power supply, power takeoff system, and feedback system. The results will be compared to the performance models, and a performance matrix will be generated.

4.5 Reporting/Cost Tracking/Administration/Scheduling (WBS 5.0)

Monthly reports and a final report will be generated to include all analysis and conclusions. The reports and test hardware will be deliverables. The schedule will be used to track and manage the effort.

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5.0 Future Research

5.1 Materials

It is anticipated that the parametric analysis and tests will identify a grouping of conductors that, when in ultrapure and thin-film form, will show a dramatic increase in drift velocity of the conducting electrons. An example of such a material would be bismuth layered with magnesium and possibly gold. It is anticipated that the magnitude of the force will be directly related to current density through the materials.

5.2 Leaky Dielectrics

It is anticipated that the parametric analysis and tests will identify a group of semiconductor materials that will have a high dielectric constant and electron mobility. An example of such material is the group III-IV semiconductors that could exhibit this effect in ultrapure form and appropriate alloy ratios. It is anticipated that the stored charge and leakage current combination will greatly enhance the force when coupled to ultrapure conductor materials. Similarly it is anticipated that the magnitude of the force will be very sensitive to current density and cross sectional area of such a combination of leaky dielectrics and conductors.

5.3 Scalability

It is anticipated that the parametric analysis and tests will identify geometric constraints on the thicknesses of the conductors and leaky dielectrics with respect to the magnitude of the force. For example, for an optimized thickness of leaky dielectric and conductor combination, the cross sectional area will be increased in order to maximize the magnitude of the force. It is anticipated that the optimal geometry will be in the form of a rectangular panel and that the magnitude of the force will scale linearly with cross sectional area of the leaky dielectric and conductor combination.

5.4 Power Generation

It is anticipated that the results of a power generation system will allow the direct development and technology implementation for a 1,000 watt or better system. The “stepping stone” demonstrator will also allow the needed test verification of independent sources. As the capacitor technology improves, the system should become smaller, lightweight, and less expensive.

5.5 Further Research

When fully developed, this system would provide propellantless transportation that is silent and leaves no emission trail. The system has few moving parts, so cost, maintenance, and risk would be greatly reduced. Vehicles would be able to travel great distances with only a small generator required. However, since this system is functional in the atmosphere, under water, and in space, the possibilities are endless. The transportation industry – airplanes, trains, and automobiles – all could be affected by this type of system.

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6.0 Required Assets

Item	Use	Comments
Variable moment arm rotor system	•Force measurements •Moment of inertia measurements •Energy Storage Tests	•Rotor arm ends are designed to accept various asymmetric capacitor geometries. •Power is communicated through bearings at top and bottom of rotor assembly.
Bifilar Pendulum	Independent measure of generated force by asymmetric capacitors	Constructed in a space frame with power communicated to the bottom of the structure via insulated guidelines.
High voltage/high current power supplies	Power supply Testing reversed polarity	•N.I.S.T. calibrated •Regulated to within 5% from 0 to 100,000 volts at a maximum current of 4 milliamps. •Polarity can be reversed
Various electrical diagnostics		oscilloscopes, volt meters, power supplies, spectrum analyzer, etc.

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7.0 ROM Cost Estimate

- Duration: 18 months
 - Materials: \$100,000
 - Coatings: \$10,000
 - Tooling \$15,000
 - Tests \$15,000
 - Hardware \$60,000
 - Labor: (3120 manhours) \$410,700
 - Principle Investigator @ \$60/hr 100% \$187,200
 - Support Engineer @ \$35/hr 100% \$109,200
 - Program Manager @ \$45/hr 75% \$105,300
 - Travel: \$ 10,000
 - Miscellaneous Equipment \$ 5,000
- TOTAL: \$516,700**